KYLE DAWSON, PHD

PhD Applied Scientist | Mathematical Optimization, Machine Learning & Geospatial Modeling

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team, setting technical direction, mathematical-modeling and evaluation standards, and mentoring scientists and engineers across geospatial machine-learning and optimization initiatives
- designed and productionized a large-scale, spatially-constrained optimization model (mixed-integer linear programming via CBC with greedy heuristics, LightGBM risk scoring on LiDAR-derived features) that allocates a fixed operational budget across a geographic infrastructure network to maximize reliability (SAIFI) per dollar — an ambiguous business problem cast as a rigorous OR formulation on GCP
- built a deployed multi-LLM / generative-AI application (Gemini on Vertex AI, Cohere North) on Cloud Run with a retrieval benchmark scoring citation accuracy against a ground-truth set, to evaluate foundation models for document Q&A and image analysis
- ship production-grade Python systems (Docker, CI/CD, BigQuery) that carry spatial and ML models from research prototype through simulation and experimentation to reliable, large-scale production

Data Scientist → Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- built and productionized machine-learning models to forecast survey quality, detect anomalies, and constrain uncertainty in gas-emission estimates, delivered through customer-facing analytics platforms (Plotly Dash on Azure) securing contracts of $50–150k
- implemented Bayesian uncertainty propagation and physics-based atmospheric boundary-layer and wind models to generate decision-grade metrics used in regulatory and enterprise reporting across 10+ countries
- invented novel measurement and quantification methodologies, presenting results to executives, customers, and international peer-reviewed conferences

Data Scientist — Ag Tech

Bayer CropScience

Jan 2020 – Mar 2022 St. Louis, MO

- applied machine learning, geospatial analytics, and weather- and soil-impact modeling to large-scale UAV- and aircraft-derived imagery for digital phenotyping and crop-performance assessment
- trained and evaluated deep-learning object-detection models (Detectron2) for classification from high-resolution georeferenced aerial imagery
- built end-to-end ML data workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets serving stakeholder decision support

Postdoctoral Fellow

NASA Langley Research Center

Feb 2018 – Jan 2020 Hampton, VA

- developed uncertainty-aware retrieval algorithms from remote-sensing observations and delivered peer-reviewed results, prototyping and experimenting with models against field data

EXECUTIVE SUMMARY

PhD applied scientist (atmospheric science) with 8+ years building mathematical-optimization, machine-learning, and geospatial modeling systems for scientifically complex, large-scale operational problems. Pairs optimization (MILP/LP) and machine learning with large-scale geospatial data (LiDAR, satellite/aerial imagery, spatial infrastructure networks), weather and atmospheric modeling, and generative AI to turn ambiguous business problems into rigorous models and shipped software. Peer-reviewed researcher who carries solutions from prototype and simulation through experimentation into reliable cloud production, and communicates results to technical and business stakeholders.

CORE COMPETENCIES

Mathematical Optimization
Operations Research
Spatial Optimization
Research & Invention
Simulation & Experimentation
Technical Leadership
Cross-Functional Collaboration
Stakeholder Communication

DOMAIN EXPERTISE

Mathematical Optimization (MILP/LP)
Geospatial Analytics
LiDAR & Satellite Imagery
Machine Learning
Deep Learning
Atmospheric & Weather Modeling
Forecasting & Time Series
Generative AI & LLMs
Anomaly Detection
Bayesian Inference & Uncertainty
Computer Vision

TOOLS & TECHNOLOGIES

Python
PuLP / CBC
PyTorch
Scikit-Learn
LightGBM
XGBoost
Detectron2
NumPy
Pandas
SQL
GCP BigQuery
Vertex AI
AWS
Azure
Docker
GPU / CUDA
MLOps / CI-CD
Git
GDAL
QGIS
Plotly
Anthropic Claude

SELECTED PROJECT

BESS Day-Ahead Arbitrage Optimizer

Self-directed proof of concept

2025

A PuLP/CBC linear program, fed by ERCOT market API data, that schedules day-ahead battery charge/discharge arbitrage against nodal settlement-point (LMP) prices under state-of-charge and power constraints.

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 — Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 — Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 — Hampton, VA

PUBLICATIONS

Google Scholar Profile